Discrete Mathematics, Algorithms and Applications
© World Scientific Publishing Company

RECOMMENDER SYSTEMS BASED ON MATRIX FACTORIZATION AND THE PROPERTIES OF INFERRED SOCIAL NETWORKS

SANTIAGO URIBE
Department of Engineering and Sciences, Pontificia Universidad Javeriana
Santiago de Cali, Colombia
suribe06@javerianacali.edu.co

CARLOS ERNESTO RAMIREZ
Department of Engineering and Sciences, Pontificia Universidad Javeriana
Santiago de Cali, Colombia
carlosovalle@javerianacali.edu.co

JORGE FINKE
Department of Engineering and Sciences, Pontificia Universidad Javeriana
Santiago de Cali, Colombia
jfinke@javerianacali.edu.co

Received Day Month Year
Revised Day Month Year
Accepted Day Month Year
Published Day Month Year

Recommender Systems are a vital part of companies with an active participation on the web. These companies require strategies that allow them to take advantage of product ratings from users in order to provide future recommendations to other users. In the last decade, several algorithms have been developed for movie recommendation, being Matrix Factorization algorithm one of the most popular. Our approach is to evaluate the performance of this recommendation algorithm in scenarios where underlying social networks, which characterize certain type of interactions between users, can be inferred. In particular, the MovieLens dataset is used, which consists of approximately one hundred thousand ratings by 671 users on 9066 movies, during the period from March 29, 1996 to September 24, 2018.

Keywords: Recommender Systems; Network Inference; Matrix Factorization; Graphs Properties.

Mathematics Subject Classification: 11xxx, 11xxx, 11xxx

1. Introduction

Due to the ever-increasing volume of online information, recommender systems have become a widely used tool by many companies, to facilitate finding relevant information to satisfy the needs and interests of users, and increase their satisfaction levels while interacting with different brands and services.

Recommender Systems (RS) make it easier for companies to limit product suggestions to only the most relevant and/or with the highest impact. These systems are algorithms that generate recommendations from product or user meta-data, to help people in their decision-making processes.

Many RS implementations suggest all kinds of products to users. Today, they are found on most company platforms with a high share of web traffic and are used to recommend music on Spotify, various products on Amazon, videos on YouTube, and even research articles to the scientific community [1]. Also, streaming platforms (such as Netflix, Amazon Prime Video, and HBO) take advantage of a variety of data sets and Machine Learning techniques to generate recommendations.

In recent years, many algorithms have been developed for movie recommendations, with great success [2]. Collaborative filtering is one of the most used approaches in RS; these algorithms operate on the principle of finding patterns in consumer behavior data, assuming if users have rated items similarly in the past, they are likely to rate similar items similarly in the future. Even though traditional collaborative filtering models generate promising results in different scenarios, they assume that users are independent and identically distributed; this assumption ignores the social interactions or connections among users. Therefore, don't explicitly use hidden patterns on users data in social networks that could add value to the prediction of such ratings.

Inferring the underlying social networks by a set of users allows us to formally characterize how users are connected. As an example, this is useful to determine how information about a certain product is disseminated among users, and in turn, allows to predict recommending of new products to users or maximizing sales of a specific product.

Sinha *et al.* [3] shows given a choice between recommendations from the network of friends and those from users in general, recommendations from friends are preferred, in terms of quality and usefulness. Consequently, RS that ignores the structure of the users social network has a more limited predictive power.

The objective of this paper is to explore the benefit of inferring an implicit social network in a dataset and using the properties of this network in a recommendation framework using the Matrix Factorization algorithm. The rest of the article is organized as follows. Section 2 is devoted to related work. Section 3, introduces the original NETINF algorithm for inferring social networks. In section 4 we talk about how matrix factoring works in item recommendation. Experimental evaluation using the MovieLens dataset is shown in Section 5. In Section 6 we present our results and discuss them in Section 7.

2. Related Work

In 2012, Gomez-Rodriguez *et al.* [4] proposed that behind data sets there are unknown underlying links, over which the information spreads. By propagating over the links, a cascade is generated and, thanks to these, together with other data, social networks can be inferred. The algorithm developed by the authors is called NETINF.

Fan *et al.* extended NETINF by merging the work of Gomez-Rodriguez, along with movie rating data, to evaluate the inference algorithm against the recommendations [5]. The authors then suggested evaluating the algorithm of Gomez-Rodriguez, on a dataset without a social network, such as MovieLens, to compare the accuracy of the recommendations.

Alpay *et al.* [12] implemented the FASTINF algorithm based on NETINF, for use in a big data environment (Map-Reduce framework) in the industry of large online social networks such as Facebook or Twitter. The generative probabilistic model used by Alpay *et al.* is similar to the work of [4], but instead of considering the cascade structure as a DAG and use the strategy of maximum weighted directed spanning tree, they consider cascade as a sparse edge set in a graph, and thus work on inferring the edges instead of the tree.

A different approach is given by Wang *et al.* [13], who considers the network inference as a classification problem and uses supervised learning to find the coefficients of a linear model, which is used as the probability of an edge existing between two nodes.

Gray *et al.* [17] use the Independent Cascade Model [10] to develop a Bayesian inference method based on Markov Chain Monte Carlo, using samples from the distribution of networks with the dynamics observed and accounting for the inherent uncertain-ties that arise in inference and data observation.

In 2017 Kose *et al.* compared the algorithms of User-Based Collaborative Filtering, Matrix Factorization, and Yehuda Koren's Integrated Model, to predict the ratings that users would give to movies [2]. This investigation resulted in a similar performance for all three algorithms, when comparing their root mean square error, or RMSE. It should be noted that the Matrix Factorization algorithm stands out due to its good results and its simple implementation.

Ma *et al.* [14] on the basis that a user's social network will affect personal behaviors on the web, present a social recommendation framework, which merges both user's social network information and rating records, using probabilistic matrix factorization.

Community-based recommendation consists of techniques that explore the network structures and social cues to perform recommendations. Gasparetti *et al.* [16] assume that the groups of users that have more similar social characteristics (even if does not have any direct relationship between them) may provide precious and additional evidence in the recommendation process.

He *et al.* [15] propose a social network-based recommender system (SNRS). This

system uses information from social networks, including user preferences, general acceptance of articles, and the social influence of friends. From this information, they develop a probabilistic model, which allows them to make personalized recommendations.

Local Trust Network (LTN) based recommendation method proposed by Liu *et al.* [18] posit that user social networks and user opinions can be incorporated to improve prediction accuracy in the domain of movie recommendation. For this they extract latent social network patterns and multiple sources of user opinions, to generate the LTN of users and to be able to perform prediction based on this network. The LTN based algorithm leads a reduction in the $RMSE$ and MAE compared to a collaborative filtering recommendation.

3. Network Inference Problem

Gomez-Rodriguez *et al.* present how to infer a network based on a generative probabilistic model and heuristics. Down below, the Cascade Transmission Model described in [4] is presented.

Given a hidden directed network G, there are multiple contagions spreading over it. As the contagion c propagates over the network, it leaves a trace, a cascade. For a set C of cascades propagating over a graph G, each cascade $c \in C$ is define as a set of tuples $(v, t_v)_c$, which means that cascade c, created by a contagion, hits (infects) a particular node v at time t_v, but it is not known how the cascade reached the node. Note that if $t_v = \infty$ it means that the node v is not reached by the cascade c. The infections are binary, i.e. a node is either infected or not.

Gomez-Rodriguez *et al.* use the Independent Cascade Model proposed by Tardos *et al.* [10], which posits that an infected node infects each of its neighbors in network G independently at random with some small chosen probability. The model implicitly assumes that each node v in the cascade c is infected by at most one node u, i.e., it does not take into account infections that come later. Thus, the structure of a cascade is completely described by a tree $T = (V_T, E_T)$, contained in the graph G.

Let $\widehat{G}$ be a hidden graph over which a set C of cascades spread over it. In order to infer this graph, first specify the probability $P_c(u, v)$ that node u spreads the contagion, c, to node v. Define $f(t_i|t_j, \alpha_{j,i})$ as the conditional likelihood of transmission between a node j and node i. The transmission likelihood depends on the infection times (t_j, t_i) and a pairwise transmission rate $\alpha_{j,i} \geq 0$. A node cannot be infected by a node infected later in time. In other words, a node j that has been infected at a time t_j may infect a node i at a time t_i only if $t_j < t_i$. For this, the authors uses three well-known parametric models: exponential, power-law and Rayleigh (see Table 1). Without loss of generality we consider $\alpha_{j,i} = \alpha$, where α is the same transmission rate for all the nodes and t_i, t_j hit time from cascades [19].

Table 1: $P_c(u, v)$ transmission models.

Model	**Transmission likelihood** $f(t_i \mid t_j, \alpha)$
Exponential	$\begin{cases} \alpha e^{-\alpha(t_i - t_j)} & \text{If } t_j < t_i \\ 0 & \text{otherwise} \end{cases}$
Power-law	$\begin{cases} \alpha(t_i - t_j)^{-1-\alpha} & \text{If } t_j + 1 < t_i \\ 0 & \text{otherwise} \end{cases}$
Rayleigh	$\begin{cases} \alpha(t_i - t_j)e^{-\frac{1}{2}\alpha(t_i - t_j)^2} & \text{If } t_j < t_i \\ 0 & \text{otherwise} \end{cases}$

Next calculate the likelihood $P(c|T)$ that cascade c in a graph G propagated in a particular tree pattern $T = (V_T, E_T)$. The cascade c propagates with probability β and stops with probability $1 - \beta$ over the edges of the tree T, therefore:

$$P(c|T) = \prod_{(u,v)\in E_T} \beta P_c(u, v) \prod_{u\in V_T,(u,x)\in E\setminus E_T} (1 - \beta) \tag{3.1}$$

Assuming independence between edges, $P(c|T)$ can be rewritten as:

$$P(c|T) = \beta^q (1 - \beta)^r \prod_{(u,v)\in E_T} P_c(u, v) \tag{3.2}$$

where $q = |E_T| = |V_T| - 1$ is the number of edges in T and it counts the edges over which the contagion successfully propagated. Similarly, r counts the number of edges that did not activate and failed to transmit the contagion: $r = \sum_{u\in V_T} d_{out}(u) - q$, and $d_{out}(u)$ is the out-degree of node u in graph G.

Now, defines $P(c|G)$, the probability that a cascade c occurs in a graph G. This can be calculated based on $P(c|T)$. We search for all possible tree structures $T \in \mathcal{T}_c(G)$ where the cascade c can propagate. Here, $\mathcal{T}_c(G)$ is the set of all the directed connected spanning trees on a subgraph of G induced by the nodes that got hit by cascade c.

$$P(c|G) = \sum_{T\in\mathcal{T}_c(G)} P(c|T)P(T|G) \tag{3.3}$$

Note that even though the sum ranges over all possible spanning trees $T \in \mathcal{T}_c(G)$, in case T is inconsistent with the observed data, then $P(c|T) = 0$. Assuming that all trees are a priori equally likely ($P(T|G) = 1/|\mathcal{T}_c(G)|$) and using the observation from Equation 3.2, we obtain:

$$P(c|G) \propto \sum_{T\in\mathcal{T}_c(G)} \prod_{(u,v)\in E_T} P_c(u, v) \tag{3.4}$$

Finally, the probability that a set of cascades C occurs in a graph G is defined as:

$$P(C|G) = \prod_{c\in C} P(c|G) \tag{3.5}$$

The Equation 3.5 defines the Cascade Transmission Model. Under this model, we seek to estimate a graph $\widehat{G}$ that (approximately) maximizes the probability that C occurs in it, i.e., to find $\widehat{G}$ that solves the following optimization problem:

$$\widehat{G} = \arg\max_{|G| \leq e} P(C|G) \tag{3.6}$$

where the maximization is over all directed graphs G of at most e edges, and $P(C|G)$ is defined by Equation 3.5.

The optimization problem seems like an intractable problem. To evaluate Eq. 3.5 is necessary to compute Eq. 3.4 for each cascade $c \in C$ on all possible trees $T \in \mathcal{T}_c(G)$ and the number of trees can be exponential in the size of G. The authors performed the Eq. 3.4 in polynomial time considering only the most likely propagation tree and applying Kirchhoff's Matrix Tree Theorem [11]. In addition, the authors introduce a concept they define as "ε-edges", which refers to the fact that nodes can become infected for other reasons than network influence. Technical details can be found in [4].

4. Matrix Factorization

Let $U = \{u_1, u_2, ..., u_m\}$ a set of m users, $I = \{i_1, i_2, ..., i_n\}$ a set of n items (in this case movies) and let T a given finite set:

$$T \subset U \times I \times X$$

with $X \subset \mathbb{R}$. The set T defines all known ratings that users gave to the movies. Every triple $(u, i, r) \in T$, with $u \in U$, $i \in I$ and $r \in X$; represents that user u rated the item i with value r. Note that typically $|T| < m \cdot n$, because each user rates only a few items. Now, let $\mathbf{R} \in \mathbb{R}^{m \times n}$ a partially specified matrix that represents set T, where each entry r_{ui} stores the rating of user u for item i.

Given a user $u \in U$ and an item $i \in I$ of which r_{ui} is unknown, the objective is to predict the rating of u on item i. For this, matrix factorization techniques are employed to learn the latent factors of users and items, to predict the unknown ratings [7]. As Cortes presents [8], recommendations can be made using Collective Matrix Factorization (CMF) models, which consider additional information about users or items.

First we talk about the low-rank matrix factorization, the goal of these MF techniques is to approximate $\mathbf{R}$ as a product of two lower-dimensional matrices:

$$\mathbf{R} \approx \mathbf{PQ} \tag{4.1}$$

where $\mathbf{P} \in \mathbb{R}^{m \times k}$ and $\mathbf{Q} \in \mathbb{R}^{k \times n}$, and k corresponds to certain latent factors.

The unknown entries of $\mathbf{R}$ cannot be represented by zero. For this case, the approximation task can be defined as follows. Let p_{uk} denote the elements of $\mathbf{P}$, and q_{ki} denote the elements of $\mathbf{Q}$. Let p_u denote a column of $\mathbf{P}$, and q_i a row of $\mathbf{Q}$.

Then:

$$\widehat{r}_{ui} = p_u \cdot q_i = \sum_{d=1}^{k} p_{ud} \cdot q_{di}$$

$$e_{ui} = r_{ui} - \widehat{r}_{ui}$$

$$SSE = \sum_{(u,i,r)\in T} e_{ui}^2 = \sum_{(u,i,r)\in T} \left(r_{ui} - \sum_{d=1}^{k} p_{ud} \cdot q_{di} \right)^2$$

$$RMSE = \sqrt{SSE/|T|}$$

$$(\mathbf{P}^*, \mathbf{Q}^*) = \underset{(\mathbf{P},\mathbf{Q})}{\arg\min}\, RMSE \tag{4.2}$$

Here $\widehat{r}_{ui}$ denotes how the u-th user would rate the i-th item, e_{ui} denotes the training error on the (u, i)-th rating, SSE denotes the sum of squared training errors. Eq. (4.2) states that the optimal $\mathbf{P}$ and $\mathbf{Q}$ minimizes the sum of squared errors only on the known elements of $\mathbf{R}$.

Now, the CMF factorizes $\mathbf{R}$ matrix together with the user attribute matrix and/or the item attribute matrix (in this case only the user attribute matrix will be used). Let $\mathbf{A} \in \mathbb{R}^{m\times s}$ user attribute matrix. Here, a_{ul} denotes the value of user u's attribute l. Let $\mathbf{C} \in \mathbb{R}^{k\times s}$, the goal its to approximate $\mathbf{A}$ as a product of two lower-dimensional matrices $\mathbf{A} \approx \mathbf{PC}$, then the equation (4.2) becomes:

$$SSE = \sum_{(u,i,r)\in T} \left(r_{ui} - \sum_{d=1}^{k} p_{ud} \cdot q_{di} \right)^2 + \sum_{(u,l)\in A} \left(a_{ul} - \sum_{d=1}^{k} p_{ud} \cdot c_{dl} \right)^2$$

$$RMSE = \sqrt{SSE/|T|}$$

$$(\mathbf{P}^*, \mathbf{Q}^*, \mathbf{C}^*) = \underset{(\mathbf{P},\mathbf{Q},\mathbf{C})}{\arg\min}\, RMSE \tag{4.3}$$

Koren *et al.* [9] showed that these models are highly prone to over-fitting if estimated as is, so they are often estimated with regularization parameters. Regularization helps to prevent over-fitting by adding a penalty term to the objective function that is being optimized. This penalty term discourages large values for the model parameters and encourages them to be closer to zero. Two commonly used types of regularization are L1 and L2 regularization. In matrix factorization algorithms, the matrices $\mathbf{P}$, $\mathbf{Q}$ and $\mathbf{C}$ are typically formed through an iterative optimization process, where the values in each matrix are adjusted to minimize the difference between the predicted and actual values in the original matrix. This optimization process is often performed using techniques such as gradient descent or

alternating least squares, with the regularization parameter controlling the balance between fitting the training data and preventing over-fitting.

5. Numerical Experimentation

5.1. *Data*

The MovieLens dataset consists of a total of 100,004 movie ratings by 671 users for 9,066 movies. These ratings can be represented by a sparse matrix, as shown in Fig. 1.

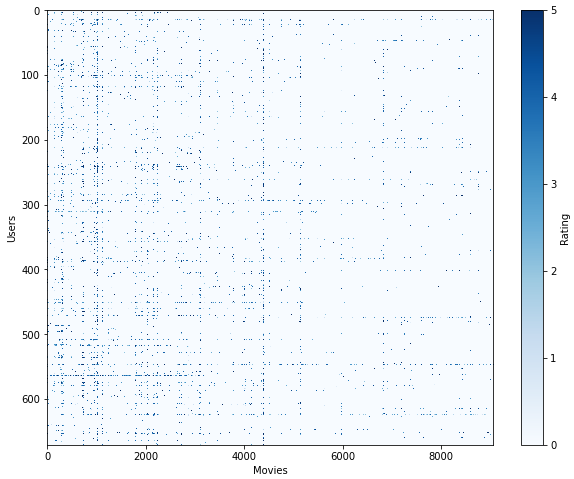


Fig. 1: MovieLens Rating Matrix

When there is a new user and there are relatively few ratings per movie or simply no ratings for this user, it is possible that he/she will not receive good recommendations. However, our proposed method infers a social network among users based on their ratings, and one solution is to add the new user as a new node in the inferred network. Different methods can be used for adding new nodes to a network, such as Barabasi's preferential attachment mechanism that adds new nodes based on the degree distribution of existing nodes, where nodes with more connections (hub nodes) are more likely to receive new connections [20]. Another phenomenon that has been observed in social networks is homophily, where individuals who share similar characteristics tend to connect with each other more often [21,22,23]. In other words, new nodes would generate connections to nodes with similar attributes. There are also methods that use both popularity and similarity in growing networks [24]. These references provide examples of how new nodes can be added to a network, and our proposed method can be adapted to incorporate these techniques to provide good recommendations even for new users.

In terms of recommendation quality, warm-start recommendations (which utilize interactions and preferences of previous users) typically outperform cold-start recommendations (which are provided to new users with limited interaction history). Nevertheless, Cortes [8] has demonstrated that the Collaborative Matrix Factorization (CMF) algorithm can produce superior results compared to other recommendation models that generate random recommendations or non-personalized "most popular" recommendations, which are based solely on the average rating of items.

To facilitate a comparison between the recommendation technique proposed in this paper and a common collaborative filtering, and able to assess the impact of inferred social networks and their properties on the quality of recommendations, we conducted a standard collaborative filtering analysis on the dataset and obtained relevant performance metrics, as presented in Table 2.

Table 2: Metrics for Movie Recommendation.

Metric	**Mean (μ)**	**Std. Deviation (σ)**
$RMSE$	0.8983	0.1867
MAE	0.7126	0.0427
R^2	0.1964	0.1715

5.2. *Network Inference*

With the MovieLens dataset, it is possible to know when a user rates a movie, but it is not known who or what influences the user to rate that movie. Thus, the NETINF algorithm takes on great importance in this area. The problem is then to infer the underlying social network of users over which the cascades propagate, in this regard, we use the analysis of movie cascades proposed by Fan *et al.* [5], where all ratings of a movie will be treated as a cascade, for a total of 9066 cascades. Each cascade c contains all the ratings r_i for a particular movie m_c, such rating can be represented in the form $(u_i, t_i)_c$ which means that the user u_i rates the movie m_c at time t_i.

Due to the different transmission functions (Table 1) wherewith NETINF infers pairwise transmission, Fig. 2 shows the inferred edges obtained.

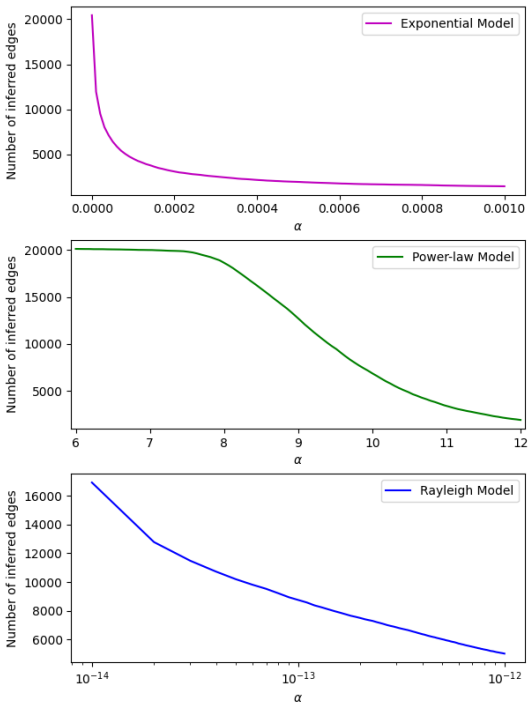
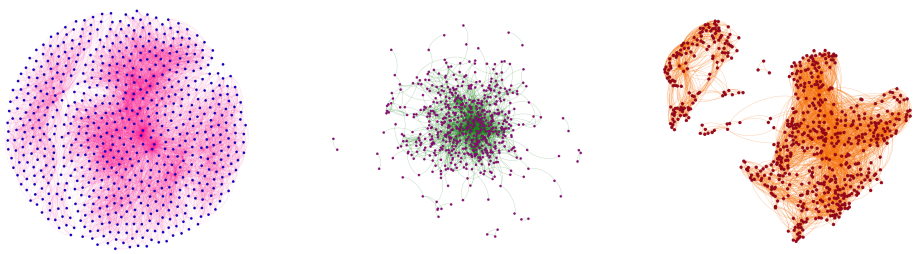


Fig. 2: Inferred Networks from NETINF

In the direction of visually appreciating the networks inferred by the NETINF algorithm based on pairwise transmission models, three different types of networks were plotted using Gephi software, as shown in Fig. 3.



(a) (b) (c)

Fig. 3: Different inferred networks. They are: (a) Exponential model, $\alpha = 1,02 \cdot 10^{-5}$ and Fruchterman-Reingold layout. (b) Power-Law model, $\alpha = 11,\overline{15}$ and Yifan Hu Proportional layout. (c) Rayleigh model, $\alpha = 3,7 \cdot 10^{-13}$ and Force Atlas layout.

6. Results

To incorporate the inferred network properties, the `SNAP` [6] library was used to extract various graph properties, including degree, Betweenness, Closeness, Eigenvector, Farness Centrality, Community Structure, Node Eccentricity, and Hubs and Authorities score. These properties were then added as user attributes to the dataset, which was used to make movie recommendations using the CMF algorithm.

Table 3: Metrics for Movie Recommendation Using Inferred Social Network Properties

Model	**Metric**	**Mean (μ)**	**Std. Deviation (σ)**
Exponential	$RMSE$	0.7867	0.0151
	MAE	0.6833	0.0074
	R^2	0.2959	0.010
Power-law	$RMSE$	0.7887	0.0171
	MAE	0.6842	0.0083
	R^2	0.2948	0.0106
Rayleigh	$RMSE$	0.7888	0.0164
	MAE	0.6840	0.0078
	R^2	0.2940	0.0113

Based on the results in Table 2, it can be observed that the common collaborative filtering technique had an average $RMSE$ of 0.8983, an average MAE of 0.7126, and an average R^2 of 0.1964. These metrics were used as a baseline to compare the performance of the system using inferred social network properties.

Table 3 shows the results obtained by applying the CMF algorithm using the dataset with added network properties inferred from the three different models. The results show that all three models outperformed the common collaborative filtering technique in terms of $RMSE$, MAE, and R^2.

It is worth noting that the Exponential model had the lowest average RMSE (0.7867) and MAE (0.6833), as well as the highest average R^2 (0.2959), suggesting that this model provided the best performance overall. However, the differences between the models are relatively small.

Additionally, a figure of the $RMSE$ values for each of the 100 models per network type was generated to visualize the distribution of performance. As shown in Fig. 4, the Exponential model had the most consistent performance across the different models, with a smaller range of $RMSE$ values compared to the Power-law and Rayleigh models. This reinforces the previous conclusion that the Exponential model provided the best overall performance, but also highlights the importance of evaluating the consistency of the results. Further statistical analysis could be conducted to determine if the observed differences between the models are significant.

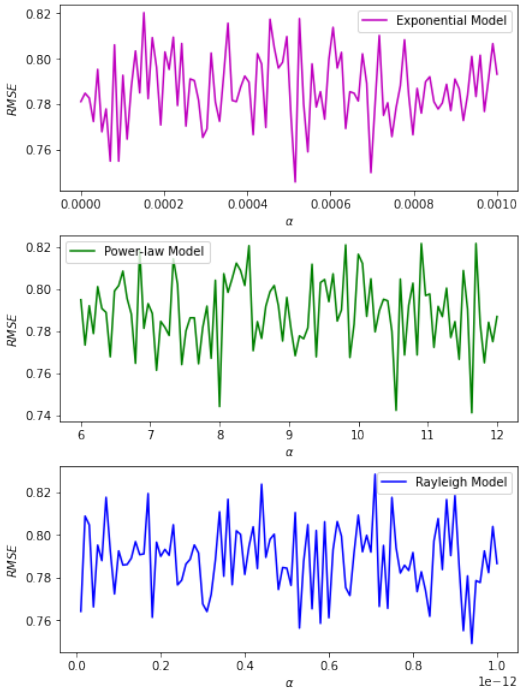


Fig. 4: $RMSE$ for Movie Recommendation with Graph Properties.

Figure 3 displays the three network layouts used for each of the three models in the experiment. At first glance, there doesn't appear to be a significant difference between the layouts and the results presented in Table 3 and Figure 4. In essence, the layouts are based on a graph representation as a mechanical system, where the edges act as springs. The aim is to minimize the energy of the system by balancing attraction and repulsion between nodes through the connecting springs, resulting in an optimal visualization of the graph. Despite the relative similarity between the three networks, they exhibit some distinctive features, such as a circular border comprising of some boundary nodes and two large components connected by hub nodes. Interestingly, network layout (a) in Figure 3 appears to be more stable and fluid, aligning with the findings of Figure 4 and Table 3.

6.1. *Performance Comparison with Benchmarks*

We selected the 11 benchmark algorithms presented in Table 3 of [25] to compare our method, Matrix Factorization with Properties of Inferred Networks (MAFPIN) using Exponential Model, including:

- **UCMF**: UCMF (User Clustering Matrix Factorization) is a graphical model that clusters users into K groups for recommendation.
- **ICMF**: ICMF (Item Clustering Matrix Factorization) is a graphical model that clusters users into K groups for recommendation, as opposite to the UCMF recommender.
- **PMF**: PMF (Probabilistic Matrix Factorization) is a probabilistic model that generates user and item latent factor vectors from Gaussian distributions.
- **BPMF**: BPMF (Bayesian Probabilistic Matrix Factorization) places priors on the user and item latent factor vectors in the Probabilistic Matrix Factorization model.
- **BMF**: BMF includes user and item biases into the matrix factorization model. The addition of bias improves the accuracy of prediction.
- **NMF**: NMF is another matrix factorization scheme that requires the latent factor vectors to be non-negative.
- **SVD++**: SVD++ is a model that merges latent factor model and neighborhood effect.
- **LMF**: LMF (Localized Matrix Factorization) first decomposes the matrix into several blocks and then conducts matrix factorization in each submatrix.
- **BCC**: The BCC (Bayesian Co-Clustering) algorithm co-clusters the matrix into several blocks so that the entries inside the same cluster have low variance.
- **LLORMA**: LLORMA (Local Low-Rank Matrix Approximation) is a state-of-the-art matrix factorization model based on the local low-rank assumption.
- **RIPC**: Implicit Preference Community coordinated Recommendation Algorithm.
- **MAFPIN**: Matrix Factorization with Properties of Inferred Networks proposed in this article.

We used *RMSE* for the assessment of rating prediction. In Table 4 we present the results along with a column (Impr.) to see the Improvement of MAPINF on others.

Table 4: *RMSE* on Movielens Dataset and Performance Comparison

Methods	*RMSE*	Impr.
UCMF	1.042	24.56%
ICMF	1.025	23.31%
PMF	0.960	18.12%
BPMF	0.954	17.61%
BMF	0.916	14.19%
NMF	0.914	14.00%
SVD++	0.912	13.81%
LMF	0.910	13.62%
BCC	1.062	25.98%
LLORMA	0.894	12.08%
RIPC	0.890	11.68%
MAFPIN	0.786	-

Overall, our method MAFPIN using Exponential Model outperforms all 11 benchmark algorithms on the Movielens dataset, achieving an *RMSE* of 0.786. The significant improvement in accuracy compared to the state-of-the-art matrix factorization models demonstrates the effectiveness of our proposed approach. The results also suggest that incorporating properties of inferred networks in matrix factorization can lead to better recommendations. Our method has the potential to be applied to a wide range of recommendation systems, especially in scenarios where user-item interaction data is sparse.

7. Conclusions and Future Work

This paper develops a technique for making recommendations (in this case of movies) using inferred social networks and the properties they present. For the development we used two algorithms, NETINF and Matrix Factorization, for the inference of social networks and the recommendation, respectively. As can be seen in Fig. 4, the technique proposed generates better results than a common collaborative filtering recommendation (only using matrix factorization algorithms), the exponential model for $P_c(u, v)$ is the one that generates the best and most stable results in the recommendation.

The authors of [4] developed a greedy algorithm to solve the network inference problem. Based on [10], they first formalize the problem in an optimization framework, as seen in equation (3.6), and show that it is NP-Hard by reduction to the Maximum Coverage Problem (*MAX-k-COVER*). They then show that the problem satisfies submodularity, so that a near-optimal solution (of at least a constant fraction $(1 - 1/e) \approx 63.21\%$ of the optimal value achievable employing k edges) can be found, using a greedy algorithm (technical details can be found in [4]).

Since the analysis of spatial and temporal complexity is highly dependent on the structure of the network and the assumptions made about it, the authors do not perform the corresponding analyses. However, the NETINF algorithm lends itself very well to parallelization, as the likelihoods of individual cascades can simply be computed independently (the NETINF algorithm in SNAP library has a C++ parallel implementation).

In the social network inference process due to the different transmission models (Table 1), different values for the transmission rate (α) were used to obtain different social network models, with a different number of inferred edges (Fig. 2); intending to explore several graphs that best model the underlying structure where movie recommendations are propagated among the users of the MovieLens dataset.

There are several interesting directions for future work. It is possible to explore other alternative applications of these recommendation models in contexts other than social networks, for example, crystalline material dynamics, where it would be interesting to know if this technique suggests connections between atoms, which would be predicted by classical models in materials physics as First Principles.

References

[1] Y. Fang and L. Si, "Matrix Co-factorization for Recommendation with Rich Side Information and Implicit Feedback," in Proceedings of the 2nd International Workshop on Information Heterogeneity and Fusion in Recommender Systems - HetRec '11, Oct. 2011, pp. 65–69. [Online]. Available: http://www.nanohub.org.

[2] A. Kose, C. Kanbak, and N. Evirgen, "Performance comparison of algorithms for movie rating estimation," in Proceedings - 16th IEEE International Conference on Machine Learning and Applications, ICMLA 2017, 2017, pp. 955–959. doi: 10.1109/ICMLA.2017.00-30.

[3] R. R. Sinha and K. Swearingen, "Comparing Recommendations Made by Online Systems and Friends," Jun. 2001.

[4] M. Gomez-Rodriguez, J. Leskovec, and A. Krause, "Inferring networks of diffusion and influence," in ACM Transactions on Knowledge Discovery from Data, Feb. 2012, vol. 5, no. 4, pp. 1–37. doi: 10.1145/2086737.2086741.

[5] C. Fan and L. Yu, "Inferring Social Networks Based on Movie Rating Data," Stanford University, CA, techreport, 2011.

[6] J. Leskovec and R. Sosič, "SNAP: A general-purpose network analysis and graph-mining library," ACM Transactions on Intelligent Systems and Technology, vol. 8, no. 1, pp. 1–20, Jul. 2016, doi: 10.1145/2898361.

[7] G. Takács, I. Pilászy, B. Németh, and D. Tikk, "Matrix factorization and neighbor based algorithms for the netflix prize problem," in RecSys'08: Proceedings of the 2008 ACM Conference on Recommender Systems, 2008, pp. 267–274. doi: 10.1145/1454008.1454049.

[8] D. Cortes, "Cold-start recommendations in Collective Matrix Factorization," ArXiv, vol. abs/1809.00366, 2018.

[9] Y. Koren, R. Bell, and C. Volinsky, "Matrix factorization techniques for recommender systems," Computer, vol. 42, no. 8, pp. 30–37, 2009, doi: 10.1109/MC.2009.263.

[10] E. Tardos, D. Kempe, and J. Kleinberg, "Maximizing the spread of influence through a social network," Proceedings of the ACM SIGKDD International

Conference on Knowledge Discovery and Data Mining, pp. 137–146, 2003, doi: 10.1145/956750.956769.

[11] W. T. Tutte, "The dissection of equilateral triangles into equilateral triangles," Mathematical Proceedings of the Cambridge Philosophical Society, vol. 44, no. 4, pp. 463–482, 1948, doi: 10.1017/S030500410002449X.

[12] A. Alpay, D. Demir, and J. Yang, "FastInf: A Fast Algorithm to Infer Social Networks from Cascades," Stanford University, CA, techreport, 2011.

[13] D. Wang, Y. Wu, and Y. Zhang, "Two Models for Inferring Network Structure from Cascades," The International Conference on Internet Computing, Apr. 2012.

[14] H. Ma, H. Yang, M. R. Lyu, and I. King, "SoRec: Social recommendation using probabilistic matrix factorization," in International Conference on Information and Knowledge Management, Proceedings, 2008, pp. 931–940. doi: 10.1145/1458082.1458205.

[15] J. He and W. Chu, "A social network-based recommender system (SNRS)," in Data Mining for Social Network Data, vol. 12, Boston, Massachusetts, USA, 2010, pp. 47–74. doi: 10.1007/978-1-4419-6287-4-4.

[16] F. Gasparetti, G. Sansonetti, and A. Micarelli, "Community detection in social recommender systems: a survey," Applied Intelligence, pp. 1–21, Nov. 2020, doi: 10.1007/s10489-020-01962-3.

[17] C. Gray, L. Mitchell, and M. Roughan, "Bayesian Inference of Network Structure from Information Cascades," IEEE Transactions on Signal and Information Processing over Networks, vol. 6, pp. 371–381, Apr. 2020, doi: 10.1109/TSIPN.2020.2990276.

[18] B. Liu and Z. Yuan, "Incorporating Social Networks and User Opinions for Collaborative Recommendation: Local Trust Network based Method," 2010. [Online]. Available: http://www.last.fm.

[19] M. Gomez-Rodriguez, D. Balduzzi, and B. Schölkopf, "Uncovering the temporal dynamics of diffusion networks," in Proceedings of the 28th International Conference on International Conference on Machine Learning, ICML'11, (Madison, WI, USA), p. 561–568, Omnipress, 2011.

[20] A.-L. Barabasi and R. Albert, "Emergence of Scaling in Random Networks", Science, vol. 286, pp. 509–512, 11 1999.

[21] M. McPherson, L. Smith-Lovin, and J. M. Cook, "Birds of a Feather: Homophily in Social Networks", Annual Review of Sociology, vol. 27, no. 1, pp. 415–444, 2003.

[22] Ö. Simsek and D. D. Jensen, "Navigating networks by using homophily and degree", Proceedings of the National Academy of Sciences, vol. 105, pp. 12758–12762, 9 2008.

[23] K. Z. Khanam, G. Srivastava, and V. Mago, "The Homophily Principle in Social Network Analysis", Proc. ACM Meas. Anal. Comput. Syst, vol. 37, p. 28, 08 2020.

[24] F. Papadopoulos, M. Kitsak, M. Á. Serrano, M. Boguñá, and D. Krioukov, "Popularity versus similarity in growing networks", Nature, vol. 489, pp. 537–540, 9 2012.

[25] X. Lin, M. Zhang, Y. Liu, and S. Ma, "Enhancing Personalized Recommendation by Implicit Preference Communities Modeling", ACM Trans. Inf. Syst., vol. 37, no. 4, Nov. 2019.